Federal Board HSSC-I Examination
Physics Model Question Paper

National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

Time allowed: 1 hour 20 minutes **Total Marks: 40**

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**. Write your answers on the allotted/given spaces.

SECTION – B (7×4 = 28 Marks)

Note: Give brief answers of the following questions. All questions carry equal marks. (7x4 = 28)

Q #	Question	Marks		Question	Marks
1	Using commutative property, distinguish between scalar product and vector product of two vectors. Give mathematical justifications for both.	2+2	**OR**	Distinguish between inertial and non-inertial frames of reference Providing at least one example for each.	3+1
2	An aero plane while flying horizontally drops a bomb when reaches exactly above the target but misses it. Explain why?	04	**OR**	Explain how inspectors can easily check the reliability of a concrete bridge with carbon fibers?	04
3	Enlist the assumptions made about an ideal fluid.	04	**OR**	What do you mean by conservative and non-conservative forces? Give two examples of each.	2+2
4	A device draws a current of **4A** when connected to a **12V** battery. Calculate the: a) power consumed by the device. b) resistance of the device.	2+2	**OR**	A straight wire of length **0.75m** carries a current of **8A**. It is placed in a uniform magnetic field of **0.2T** such that it makes an angle of **45°** with the magnetic field lines. **(a)** Determine the magnitude of the magnetic force on the wire. **(b)** What would be the force if the wire were placed parallel to the magnetic field?	3+1
5	How beats are produced? Why sometimes beats are not heard as distinct sounds by the human ear?	2 + 2	**OR**	In what ways a refrigerator operates as a heat engine working in reverse? Draw block diagram to show the direction of heat flow and work done.	3+1
6	A metal mesh enclosure blocks external electric charges from affecting a person inside. What physical principle explains this effect and how?	1+3	**OR**	Explain how the path difference between two coherent waves determines whether they produce a bright or dark interference pattern.	2+2
7	Define baryons and mesons. Give two examples of each.	2+2	**OR**	What is drag force? Mention three factors upon which it depends?	1+3

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 63**

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly. **(13 x 3 = 39)**

(i)	Deduce the dimensions of k from Coulomb's law $\left(F = k\frac{q_1 q_2}{r^2}\right)$.	03	OR	Describe any three properties of cross product of two vectors.	1x3
(ii)	Describe photon as a force carrier.	03	OR	Distinguish between the terms elastic deformation and plastic deformation.	03
(iii)	Show that: $\vec{A}.\vec{B} = A_x B_x + A_y B_y + A_z B_z$	03	OR	Show that ε and $\frac{\Delta\phi}{\Delta t}$ have the same units.	03
(iv)	Differentiate between conservative and non-conservative forces with examples.	03	OR	Distinguish between turbulent and streamline flow.	03
(v)	Calculate the resistance of wire 10 m long that has diameter of 3mm and resistivity of $2.63\times10^{-2}\Omega m$.	03	OR	A body of mass $(27.2\pm0.1)kg$ has a volume of $(2.0\pm0.1)m^3$. Calculate the density of the body? Find uncertainty in the density measurement.	03
(vi)	Derive the relationship $P.E = \frac{1}{2}\left[\frac{YA}{L}\right]x^2$	03	OR	In thermodynamics briefly describe: Reversible process Irreversible process Cyclic process	1x3
(vii)	A 5 kg shell is fired with a muzzle velocity of $700ms^{-1}$ from the gun barrel 2.4m long. Calculate the average value of: The acceleration The time to come out of the barrel	03	OR	Describe the working of Atomizers in perfume bottles.	03
(viii)	What is meant by precision? Also write the importance of taking multiple readings in an experiment.	1+2	OR	Find the area of the parallelogram whose adjacent sides are given by the vectors: $A = (\hat{i} + 6\hat{j} + 2\hat{k})m$ and $B = (7\hat{i} + \hat{j} + 5\hat{k})m$.	03
(ix)	Briefly explain light dependent resistor (LDR). Give one application of LDR.	2+1	OR	The front wheel of a tractor travels 700 revolutions while the rear wheel 280 in a time interval of 40 seconds. Find their angular velocities.	03
(x)	Compare constructive and destructive interference.	1.5 +1.5	OR	Drive a relationship between torque and angular acceleration.	03
(xi)	State first law of thermodynamics. Briefly explain the term internal energy.	1.5 + 1.5	OR	What is meant by potential gradient? Derive its formula.	03
(xii)	Briefly explain the relationship between pressure and volume of a gas at constant temperature. Show it graphically as well.	2+1	OR	What is meant by superfluid? How is superfluid made?	1.5 + 1.5
(xiii)	Water flows through a 1cm diameter pipe with a speed of $1ms^{-1}$. What should be the diameter of the nozzle if the water is to emerge at $2.1ms^{-1}$?	03	OR	A stationary observer hears the sound of a car horn at a frequency that is 15% higher than the horn's actual frequency. If the speed of sound is $343ms^{-1}$, what is the speed of the car as it approaches the observer?	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	Derive expressions for: a. Range of projectile and b. Height of projectile	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss time dilation and length contraction. c. At what speed would the mass of the proton be tripled? The rest mass of a proton is 1.673×10^{-27} kg.	2+2 +2
Q.4	a. State Bernoulli's equation. b. Derive velocity of efflux using Torricelli's theorem. Draw diagram as well.	1+3 +2	OR	Derive an expression for a magnetic force experienced by a moving charge in a uniform magnetic field. Also discuss the direction of that force.	4+2
Q.5	a. Discuss working of a refrigerator. b. How increase in Entropy means degradation of energy? Explain with an example.	3+3	OR	a. State Kirchoff's laws for current and voltage. b. There is a copper coil having 2000 turns with $0.8mm^2$ cross-section area, the length per turn is 80cm. Calculate the resistance of the coil.	3+3
Q.6	a. Derive the expression for Kinetic energy. b. State work energy theorem in a resistive medium.	3+3	OR	Derive f_1, f_2 and f_3 for modes of vibration in a closed organ pipe.	2+2 +2

$F = G\frac{m_1 m_2}{r^2}$ $\quad \vec{A}.\vec{B} = AB\cos\theta$ $\quad KE = \frac{1}{2}mv^2$ $\quad \text{density} = \frac{\text{mass}}{\text{volume}}$ $\quad v = \sqrt{2gh}$ $\quad R = \frac{\rho L}{A}$

$PE = mgh$ $\quad \vec{A}\times\vec{B} = AB\sin\theta\hat{n}$ $\quad I = I_o\cos^2\theta$ $\quad A = \pi r^2$ $\quad P = \vec{F}.\vec{v}$ $\quad A_1v_1 = A_2v_2$

$c = 3\times10^8 ms^{-1}$ $\quad \text{Work} = \vec{F}.\vec{x}$ $\quad MV = mv$ $\quad v_f = v_i + at$ $\quad 2aS = v_f^2 - v_i^2$ $\quad v = r\omega$

$\omega = \frac{\theta}{t}$ $\quad N = \frac{S}{2\pi r}$ $\quad m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$ $\quad f_o = f_s\frac{v}{v \pm v_s}$ $\quad H = \frac{v_i^2\sin^2\theta}{2g}$ $\quad E = -\frac{\Delta V}{\Delta r}$

$\theta = 2\pi r$ $\quad \rho_{copper} = 0.02\mu\Omega m$

PHYSICS HSSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 63**

SECTION – B (Marks 39)

Q. 2 Answer the following parts briefly. **(13 x 3 = 39)**

(i)	Deduce the dimensions of universal gravitational constant (G). $\left(F = G\frac{m_1 m_2}{r^2}\right)$	03	OR	Describe any three properties of dot product of two vectors.	1x3
(ii)	Give Comparison of amorphous and polymeric solids. (Three points)	1x3	OR	State the Higgs Boson as a fundamental particle which is responsible for the particle' s mass.	03
(iii)	The magnitudes of dot and cross product of two vectors are $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.	03	OR	How a seismometer uses electromagnetic induction to detect earthquakes? Explain briefly.	03
(iv)	In a science experiment a 3.00 kg water rocket, is launched from ground. The rocket's total energy at the top of its flight is 2352 J. a. What was the rocket's launching speed? b. What height did the rocket went?	03	OR	Find the power delivered by the engine for attaining velocity (3,4) ms^{-1} while it exerts a force (8, -2) N.	03
(v)	Calculate the resistance of a wire 10 m long that has diameter of 2 mm and resistivity of $2.63\times10^{-2}\Omega m$.	03	OR	Differentiate between accuracy and precision of measurements. (Give a graphical example)	03
(vi)	Briefly explain brittle and ductile materials. (In context of elastic and plastic deformation)	1.5 +1.5	OR	State first law of thermodynamics. Discuss its significance.	1+2
(vii)	Evaluate effect of air resistance on projectile motion. Show graphically.	2+1	OR	Briefly explain Magnus effect (an application of Bernoulli Effect). Draw diagram as well.	2+1
(viii)	Write any three limitations of dimensional analysis.	03	OR	Briefly discuss any three properties of an ideal fluid.	1x3
(ix)	What is magnetic flux? Also describe magnetic flux density and write its formula.	1+2	OR	Write any three characteristics of stationary waves.	1x3
(x)	What are beats? How are they produced?	1+2	OR	Determine the equivalent value of 1 radian in degrees.	03
(xi)	Briefly describe: a. Unit cell and its types b. Crystal lattice	2+1	OR	Derive the relation between Torque and angular acceleration.	03
(xii)	What is a light dependent resistor (LDR)? Why is LDR useful in automatic circuits?	1+2	OR	Explain the use of beats in tuning musical instruments.	03
(xiii)	The orbit of an electron revolving around the nucleus of an atoms is considered to be circular of radius 50 pm and the period of motion is 0.2 fs. Find out angular velocity of electron (ω).	03	OR	The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C. Calculate the distance of the point from the charge.	03

SECTION – C (Marks 24)

Attempt the following questions.

Q.3	a. What is meant by polarization. b. Discuss one application of polarization. c. A polarized light of intensity I_o is passed through another polarizer whose axis makes an angle of 60° with the axis of the polaroid filter. What is the intensity of transmitted polarized light from second polaroid filter?	1+2 +3	OR	a. State Archimedes principle. b. A water storage tank in a residential area develops a small leak near its lower end. The top of the water in the tank is 10m above the point of the leak. (i) With what speed does the water rush from the hole? (ii) If the hole has an area of $0.080cm^2$, how much water flows out in one second?	1+5
Q.4	Describe elastics collision in one dimension. Derive relation to show the relative velocities before and after collision.	1+5	OR	Briefly explain Elastic Potential Energy. Derive its mathematical formula.	2+4
Q.5	State working principle of Carnot's Engine. With the help of graph discuss its all four processes in detail.	1+1 +4	OR	A ball of 100gm is thrown vertically upward at a speed of 25m/s. If no energy is lost, determine the height it would reach. If the ball only rises to 25m, calculate the work done against the air resistance. Also calculate the force of friction.	06
Q.6	a. What is meant by term protentional gradient? Derive its formula. b. Show that $\frac{volt}{meter} = \frac{newton}{columb}$	3+3	OR	a. State postulates of special theory of relativity. b. Briefly discuss length contraction as a consequence of special relativity. c. The length of spaceship is 100m long. What will be its length if the spaceship moves with a speed of 0.99c?	2+1 +3

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$\omega = \frac{\theta}{t}$ $\quad N = \frac{S}{2\pi r}$ $\quad m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$ $\quad f_o = f_s\frac{v}{v \pm v_s}$ $\quad H = \frac{v_i^2\sin^2\theta}{2g}$ $\quad E = -\frac{\Delta V}{\Delta r}$

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Physics Model Question Paper
Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)

Time allowed: 2.35 hour **Total Marks: 63**

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**.
Write your answers on the allotted/given spaces.

SECTION – B (Marks 39)

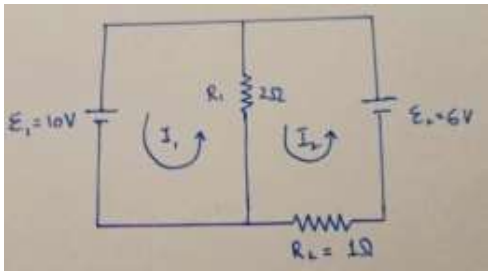
Q.2 Attempt the following questions. **(13x3 = 39)**

Q#	Question	Marks		Question	Marks
(i)	An equation is dimensionally correct. Does it mean that equation is necessary correct?	03	**OR**	Find cross product of two unit vectors at 0° and 90°.	03
(ii)	Define crystalline and amorphous solid.	03	**OR**	What are the differences between gluons and photons?	03
(iii)	Write the characteristics of scalar product.	03	**OR**	Define the principle used in seismometer.	03
(iv)	Differentiate between conservative and non-conservative forces.	03	**OR**	How a steady flow is different from turbulent flow?	03
(v)	What is effect of radius of cross section of conductor on its resistance?	03	**OR**	Radius of sphere is measured with a screw gauge and is substituted in the formulae of its surface area($A=\pi r^2$) and volume($V=4/3\pi r^3$). Which of the two results will be more accurate?	03
(vi)	Explain the stress strain curve for ductile material.	03	**OR**	Describe the limitation of first law of thermodynamics.	03
(vii)	Find the angle of projection of a projectile for which the maximum height and corresponding range are equal?	03	**OR**	An ideal liquid is flowing through a pipe of variable diameter as shown in figure. Where is pressure high (at position A or at B). Justify your answer. 	03
(viii)	By using concept of dot product, how can you describe that magnetic flux is scalar product of magnetic flux density (**B**) and vector area (**A**)?	03	**OR**	Is there any transfer of energy through a medium when stationary waves are produced in it? Explain.	03
(ix)	Two tuning forks of frequencies 456 Hz and 452 Hz are sounded together. How many beats are produced over a period of 10 second?	03	**OR**	A 40 kg mass is supported by 5 m long aluminium wire ($Y_A=7\times10^{10}$ Pa). To have same elongation in a copper wire of the same length under the same weight, find the diameter of copper wire. ($Y_C=12\times10^{10}$ Pa)	03

(x)	Why does the resistance of LDR decrease with increase in intensity of light on it?	03	**OR**	Define (a) radian (b) angular acceleration	03
(xi)	If 280 J of work is done in carrying a charge from a place where potential is -12V to another place where potential is V volt. Calculate the value of V.	03	**OR**	Write the characteristics of an ideal fluid.	03
(xii)	Two spheres A and B are of the same masses but the radius of A is greater than that of B. Which one will require more torque applied on it to bring into rotation about their axes of rotation?	03	**OR**	Hydrogen has three isotopes. How can you use magnetic field to separate them?	03
(xiii)	Define constructive and destructive interference.	03	**OR**	State the process in which the internal energy of the system remains constant?	03

SECTION – C (Marks 24) **(4x6 = 24)**

Note: Attempt all questions. Marks of each question are equal.

Q.3	Explain elastic collision in one dimension and prove that for two bodies colliding elastically, relative speed of approach before collision is equal to relative speed of separation after collision.	02+ 04	**OR**	Explain Carnot cycle. Derive the relation for the efficiency of Carnot engine.	03+03
Q.4	State law of conservation of angular momentum. Show that $\Delta L/\Delta t = 0$ and angular momentum L is conserved for an isolated system.	01+03 +02	**OR**	Show that electric field intensity is equal to negative of potential gradient. Prove that Volt/meter = Newton / Coulomb	04+02
Q.5	What is meant by quark? Explain the quark family of particle.	02+ 04	**OR**	Describe the following consequences of special theory of relativity A: Mass variation B: Length contraction C: Time dilation	02+02 +02
Q.6	Find current flowing through each resistor, using Kirchhoff's law in the given circuit. 	06	**OR**	In an experiment a 3kg water rocket is launched from ground. The rocket total energy at the top of its flight is 2352J. a. What was the rocket launching speed? b. What height did the rocket reach? c. What is the kinetic energy and potential energy of the rocket after 2.5s of its launch?	02+02 +02

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Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

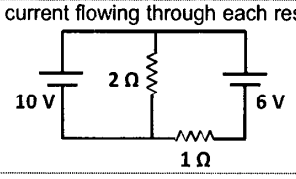
Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	Briefly explain 'precision' and 'accuracy' with a diagram.	1x3	OR	How can ordinary light be polarized using polarizers?	03
(ii)	The velocity of a missile traveling at 1500 ms^{-1} increases to 5000 ms^{-1} after moving through a distance of 70 km. Calculate the time for the missile to reach this velocity.	03	OR	A body at 200°C undergoes a reversible isothermal process. The heat energy removed in this process is 7300 J. Calculate the entropy change and discuss entropy sign.	03
(iii)	What is meant by 'conservative' and Non–conservative' forces? Give one example each.	03	OR	What is meant by Magnus effect? Explain with the help of an example.	03
(iv)	Differentiate between crystalline solids and amorphous solids. (Give at least three differences.)	03	OR	Earth completes its one rotation about its axis in 24 hours. Calculate angular velocity of earth about its axis in radian per second.	1+2
(v)	The Doppler's effect is used in radar system. Briefly explain the working of radar system.	03	OR	Verify that: $(\vec{A}.\vec{B})^2+(\vec{A}\times\vec{B})^2=(AB)^2$	03
(vi)	Calculate KE in electron volt and joules gained by an electron moving through a potential difference of 2 volts.	03	OR	What happens to the density of an object as it moves in a direction with very high speed relative to speed of light? Justify.	03
(vii)	Why does the resistance of LDR decrease with increase in intensity of light on it?	03	OR	State Faraday's law of electromagnetic induction. Give its mathematical form.	2+1
(viii)	Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit? Justify.	1.5 + 1.5	OR	At what speed would the mass of proton be doubled? The rest mass of proton is $1.673\times10^{-27}\,kg$.	03
(ix)	The Kinetic energy of a charged particle moving in a uniform magnetic field does not change. Why? Explain briefly.	03	OR	If $^{233}_{92}U$ decays twice by alpha particle emission, what is the resulting isotope? Show by equation.	03
(x)	What is Faraday's cage? Briefly explain its working.	03	OR	Compare gluons and photons.	03
(xi)	What is meant by annihilation? Give one annihilation reaction.	03	OR	To increase the intensity of wave by a factor of 10, calculate the required increase in amplitude of wave.	03
(xii)	How will the radius of a flexible ring change if it is given negative charge?	03	OR	What is the difference between inertial and non-inertial frame of reference?	03
(xiii)	Use dimensional analysis to prove the equation $P=\rho gh$ is dimensionally homogeneous.	03	OR	Write three statements of second law of thermodynamic. (one in terms of entropy)	03
(xiv)	Derive the relation between torque, moment of inertia and angular acceleration.	1x3	OR	What is meant by interference? Give conditions for interference.	1+2

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	State and explain vector product with an example. Also show that $\vec{A}\times\vec{B}\neq\vec{B}\times\vec{A}$	3+4	OR	What is meant by projectile? Derive the mathematical expressions of following for projectile motion: i) Time of flight ii) Range and maximum range of projectile	1+2 +4
Q.4	Explain the working of refrigerator. Also explain the coefficient of performance of refrigerator.	4+2	OR	State work-energy theorem. Explain this theorem for resistive medium.	06
Q.5	State the equation of continuity and derive its mathematical form. On what basis is the equation of continuity derived?	1+5 +1	OR	What is meant by elastic potential energy? Find a relation for the elastic potential energy of a material.	1+6
Q.6	Why does an astronaut feel weightless in space? How can artificial gravity be produced in a satellite? For a satellite derive expression for angular velocity, time period and frequency of satellite.	1+1 +4	OR	Find the current flowing through each resistor of the circuit. 	06

- $\sin^2\theta+\cos^2\theta=1$
- $\omega=\frac{\theta}{t}$
- $S=\frac{\Delta Q}{T}$
- $1eV=1.6\times10^{-19}J$
- $\vec{A}.\vec{B}=AB\cos\theta$
- $\sum\varepsilon=0$
- $\omega=\frac{2\pi}{T}$
- $\vec{A}\times\vec{B}=AB\sin\theta$
- $E=qv$
- $I=\frac{\varepsilon}{R}$
- $v_f=v_i+at$
- $v_f^2-v_i^2=2aS$
- $F_c=\frac{mv^2}{r}$
- $\varepsilon=\frac{\Delta\phi}{\Delta t}$
- $I\propto A^2$
- $\sum I=0$
- $m=\frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$

PHYSICS HSSC–I

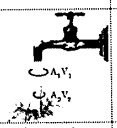
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Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

(i)	Show that the equation $P = \rho g h$ is dimensionally homogeneous.	03	OR	Explain that increase in entropy means degradation of energy.	03
(ii)	Time for 10 vibrations of a simple pendulum is 17.6s. Least count of stop watch is 0.1s. Find time period of simple pendulum. Also determine uncertainty in it.	03	OR	When water falls from a tap, its cross-sectional area decreases as it comes down. Explain why? 	03
(iii)	Under what condition the rectangular (perpendicular) components of a vector have same magnitude?	03	OR	Discuss the motion of two bodies after collision when $m_1 << m_2$ and $V_2 = 0$.	03
(iv)	What is meant by Gyroscope? Explain briefly.	03	OR	What is meant by work? How can work done be calculated graphically by a variable force?	03
(v)	A space ship has diameter of **8m** rotates about its axis to produce artificial gravity of magnitude **17.47ms^{-2}** in it. Calculate with which angular speed the satellite rotates?	03	OR	Differentiate between conservative and non-conservative forces. Give examples.	03
(vi)	Derive the relation between torque, moment of inertia and angular acceleration.	03	OR	Derive the formula for Kinetic Energy [using the equations of motion]	03
(vii)	What is meant by Magnus effect? Give its causes and effects.	03	OR	Differentiate between reversible and irreversible processes. Also give examples.	03
(viii)	Show that: $\frac{Volt}{meter} = \frac{Newton}{Coulomb}$	03	OR	Find the power delivered by an engine for attaining velocity= $(4\hat{i} + 5\hat{j})$ m/s while it exerts a force= $(8\hat{i} - 2\hat{j})$ N.	03
(ix)	What is meant by null method of measurement? Enlist two applications.	03	OR	Briefly describe the terms, *stress*, *strain* and *Young's modulus*.	03
(x)	Why are the suspension bridges written with their period of use? Why after that period it is dangerous to use the bridge? (Elastic potential)	03	OR	What is meant by magnetic flux linkage? Explain briefly.	03
(xi)	State three statements of second law of thermodynamics.(one in terms of entropy)	03	OR	How is Doppler's effect helpful in diagnosis of cardiac diseases? Explain briefly.	03
(xii)	Calculate entropy change when 800g of ice melts at $0°C$, latent heat of fusion for ice is $3.36\times10^5 J/kg$. Also discuss sign of entropy.	03	OR	Can sound waves be polarized? Explain briefly.	03
(xiii)	To increase intensity of waves by a factor of 25, calculate the required increase in amplitude of wave.	03	OR	What is super fluidity? Give some of its applications.	2+1
(xiv)	Under what conditions can the terminal potential differences of a battery exceed its emf?	03	OR	State the characteristics of vector product.	03

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	What is meant by projectile? Determine the expressions for velocity and displacement of a projectile thrown horizontally from a certain height.	1+4 +2	OR	What is meant by electric field strength? Show that: $E = \frac{-\Delta V}{\Delta r}$	07
Q.4	State and explain terminal velocity. Derive its equation by using stoke's law.	1+2 +3	OR	What is meant by quark? Explain the quark family of particles.	2+4
Q.5	What are stationary waves? Discuss the formation of harmonics in stationary waves in an open organ pipe.	1+6	OR	What is centripetal acceleration? Derive an expression for centripetal acceleration in terms of angular velocity "ω".	1+6
Q.6	Discuss and derive an expression for magnetic force acting per unit length of conductor, when current is passing through two parallel straight conductors in upward direction.	06	OR	Discuss the following consequences of special theory of relativity: i) Time dilation ii) Length contraction iii) Mass increase	2x3

- $\Delta S = \frac{\Delta Q}{T}$
- $v_2' = \left(\frac{2m_1}{m_1 + m_2}\right) v_1$
- $T = \frac{Total\ time}{No.of\ vibrations}$
- $P = \bar{F}.\bar{v}$
- $I \propto A^2$
- $\omega = \sqrt{\frac{g}{R}}$
- uncertainty $= \frac{\text{Absolute uncertainty}}{\text{No. of vibrations}}$
- $\Delta Q = mL_f$
- $v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_1$
- $W = KE = F.\Delta d$
- $F_{drag} = 6\pi\eta r v$
- $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$

PHYSICS HSSC–I

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

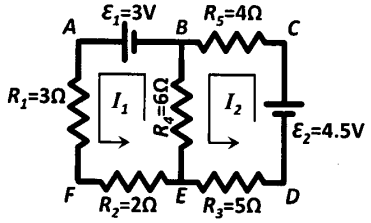
Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	Derive the relation for centripetal acceleration (a_c) of on a body moving in a circle of radius r with velocity v.	03	OR	Find the power delivered by the engine for attaining velocity (2,3) m/s, while it exerts a force (7,-3) N.	2+1
(ii)	What is meant by dot product of two vectors? Give mathematical expression and one example.	1x3	OR	Derive PV=kNT for an ideal gas.	03
(iii)	Differentiate between constructive and destructive interference of waves.	1.5+ 1.5	OR	A rabbit jumps over an object 0.7 m high: a. Considering just its vertical motion, calculate its vertical speed when it leaves the ground. b. How long a time it stays in the air?	1+2
(iv)	Briefly describe the terms Stress, Strain, and Young's modulus.	03	OR	Briefly explain the applications of Doppler Effect in radar and astronomy.	1.5+ 1.5
(v)	What is meant by elastic potential energy? Give its mathematical form.	2+1	OR	A projectile is launched horizontally from the top of a 50 m high cliff and lands at a distance of 20 m from the base of the cliff. Determine the magnitude of the launch velocity.	03
(vi)	The energy of photon is given by E=hf, find the dimensions of Plank's constant h, where f is frequency.	03	OR	A point source of sound emits sound energy at a rate of 15 W. Calculate the intensity of sound at a distance of 4 m from the source.	03
(vii)	A ball of mass 120 g is released from a height of 30 m. if the ball encounters an air resistance of 0.3 N, find the kinetic energy of the ball just before striking the ground.	03	OR	Briefly describe the phenomenon of polarization in transverse waves.	03
(viii)	Briefly describe the elastic deformation and plastic deformation.	03	OR	Calculate the electric field intensity at a distance of 0.5 m from a point charge of 2×10^{-6} C in a vacuum.	03
(ix)	A current of 2 A is flowing in a copper conductor with a cross-section of 1 mm^2. Find the drift velocity of the electrons. (For copper, n=8.5×10^{28} per m^3)	03	OR	Derive the formula for Kinetic Energy [using the equations of motion].	03
(x)	Find the efficiency of a heat engine operating between 427 °C and 27 °C.	03	OR	The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C. Calculate the distance of the point from the charge.	03
(xi)	Determine the relationship for maximum height of a projectile.	03	OR	Differentiate between conservative and non-conservative forces.	03
(xii)	An object having a net charge of 24μC is placed in a uniform electric field of 610 N/C directed vertically. What is the mass of this object if it floats in the field?	03	OR	A proton is moving in a circular orbit of radius 12 mm in a uniform 0.30 T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.	03
(xiii)	Find the momentum of a positron, moving with relativistic speed. The energy of the positron is 0.47 MeV.	03	OR	According to which principle ships float on water? Explain briefly.	03
(xiv)	Distinguish between crystalline and amorphous solids.	1.5+ 1.5	OR	Calculate the resistance of wire 15 m long that has a diameter of 1.5 mm and resistivity of 2.63×10^{-2} Ω m.	1+2

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	What is meant by angular momentum? Determine the angular momentum of a rigid body.	2+5	OR	State and explain the second law of thermodynamics. Also draw diagrams.	3+2 +2
Q.4	Explain the origin of the forces between current-carrying conductors and determine the direction of the forces.	4+2	OR	How does the frequency of sound of a source appear when it moves: a. Towards a stationary observer b. Away from a stationary observer	4+2
Q.5	Explain the working of venturi meter. Give its mathematical form. Draw its sketch.	07	OR	What is meant by quark? Explain the quark family of particle.	2+5
Q.6	Find Current flowing through each resistor, using Kirchhoff's law, in the given circuit.  	06	OR	What is special theory of relativity? Describe the following consequences of special theory of relativity: a. Mass variation b. Length contraction c. Time dilation	2+2 +2

$g = 9.8ms^{-2}$ • $V = Er$ • $P = \vec{F}.\vec{v}$ • $qE = mg$ • $A_1v_1 = A_2v_2$ • $2gh = v_f^2 - v_i^2$ • $mgh = K.E - Work\ done\ by\ air\ resistance$

• $x = v_i \times t$ • $F_e = F_g$ • $eBr = vm_p$ • $I = nAvQ$ • $v_f = v_i + gt$ • $p^2 = \frac{E^2 - m^2c^4}{c^2}$ • $k = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ Nm^2C^{-2}$

• $I = \frac{P}{A}$ • $R = \rho\frac{L}{A}$ • $E = \frac{kq}{r^2}$ • $E = \frac{F}{q}$ • $F = \frac{kq_1q_2}{r^2}$ • $y = v_it + \frac{1}{2}gt^2$ •

PHYSICS HSSC–I
(Curriculum 2022-23) (OLD)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

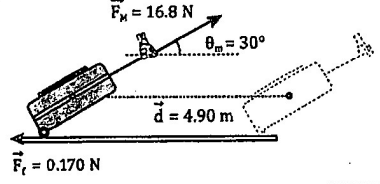
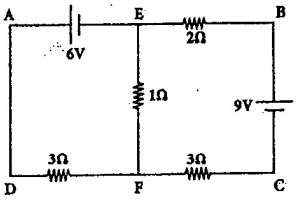
Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	If voltage measured across a conductor is $V \pm \Delta V$ = (7.3 ± 0.3) volts and current, is I±ΔI = (2.73 ± 0.051) ampere. Find the resistance and uncertainty in it?	03	OR	A peddler is pushing a trolly on a horizontal road with a force of 50 N making 30° with the road. Find the horizontal and vertical components of force?	1.5 + 1.5
(ii)	Prove that the given equation $v_f = f_i + at$ is dimensionally correct.	03	OR	If $\|A \times B\| = 3/4$ and $\|A.B\| = 1/2$ then find the angle between A and B?	03
(iii)	Derive a mathematical relation for height of projectile.	03	OR	Determine horizontal and vertical component of velocity of a projectile when launched horizontally.	03
(iv)	What is relation between linear and angular acceleration? Derive a mathematical expression.	03	OR	What is relation between Torque and Angular Momentum? Derive a mathematical expression.	1+2
(v)	Deriver the formula for kinetic energy.	03	OR	Explain work-energy theorem in a resistive medium.	1+2
(vi)	Using Archemedes principle, prove that upthrust is equal to weight of the liquid displaced. (Diagram is not required)	03	OR	Discuss the working of aerofoil.	03
(vii)	Calculate the strain energy stored in a 1m long steel wire connected to the roof, holding a weight of 980N. The diameter of steel wire is 0.004m and $\Delta L = 0.00039m$.	03	OR	A copper wire has a length of 50 m and cross-sectional area of 1.5 x 10^{-6} m^2. The resistivity of copper is 1.68 x 10^{-8} Ω m. Calculate the resistance of the wire?	03
(viii)	Using Ideal Gas equation verify that PV = NkT	1.5 +1.5	OR	Apply Kinetic Theory of Gases to prove that ρ =PM/RT	03
(ix)	If amplitude of a wave is doubled how much energy is increased?	03	OR	How is magnetic field created around a straight current carrying conductor? How its direction is determined?	2+1
(x)	An electric field has a strength of 5 N/C. What force is exerted on a charge of 2 C placed in this field?	03	OR	Prove that $E = -\frac{\Delta V}{\Delta r}$	03
(xi)	What is meant by the elastic potential energy of a material? Derive a mathematical relation for it.	1+2	OR	Using diagram derive formula for unknown resistance in Wheatstone Bridge.	1+2
(xii)	A particle with charge -3e moves at a velocity of 1 x 10^6 m/s at an angle of 45° and experiences a magnetic force of 6.72 x 10^{-14} N. What is the magnetic field strength?	03	OR	The frequency of a certain string is 125 Hz, when it vibrates in 5 segments. Calculate the frequencies when the string is vibrating in 3 and 4 segments. What is the speed of the wave in it, if its length is 0.80m?	03
(xiii)	What is Mass-Energy Equivalence Relation? Explain it briefly.	1+2	OR	At what speed would the mass of proton be tripled? The rest mass of proton is 1.673 x 10^{-27} kg.	03
(xiv)	Distinguish between α, β and γ particles on the basis of their charge and mass?	1x3	OR	State the Higgs Boson as a fundamental particle which is responsible for the particle's mass.	03

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	Explain Carnot Engine and Carnot Theorem with the help of its graph. Find the relation for its efficiency.	4+3	OR	a. Describe the Doppler's effect for a moving source and a stationary listener. b. Discuss any two applications of Doppler's effect.	4+3
Q.4	Explain how electric and magnetic fields can be used in velocity selection. Make diagram as well.	4+2	OR	State Coulomb's Law. Write its mathematical expression. Two point charges $q_1 = 4\mu C$ and $q_2 = 2\mu C$, exert a repulsive force of 36N on each other. Calculate the distance between the charges.	1+1 +4
Q.5	A girl is dragging a suitcase and pulls it a distance of 4.90 m along level ground. She is applying a constant force of 16.8N at 30° with the horizontal. A 0.170 N friction force opposes the suitcase's motion. Find (a) work done by Girl (b) work done by frictional force and (c) net work done by all the forces acting on the suitcase? 	07	OR	a. Derive the formula for combined resistance of two or more resistors in series. b. Calculate the current that flows in the 1Ω resistor in the following circuit. 	3+4
Q.6	Determine f_1, f_2 and f_3 for stationary waves in an open pipe.	2x3	OR	How is artificial gravity created to counter weightlessness. Determine linear velocity angular velocity, time period and frequency of artificial satellite.	2+4

- $V = IR$
- $I = \frac{P}{A}$
- $F = \frac{kq_1q_2}{r^2}$
- $1eV = 1.6 \times 10^{-19} J$
- $\overline{A}.\overline{B} = AB\cos\theta$
- $A = \pi r^2$
- $E = \frac{F}{q}$
- $\overline{A} \times \overline{B} = AB\sin\theta$
- $f_n = nf_1$
- $v = f\lambda$
- $v_f = v_i + at$
- $v_f^2 - v_i^2 = 2aS$
- $F_c = \frac{mv^2}{r}$
- $P.E = \frac{1}{2}\left[\frac{YA}{L}\right]x^2$
- $R = \frac{\rho L}{A}$
- $F = qrB\sin\theta$
- $Y_{steel} = 20 \times 10^{10} N/m^2$

PHYSICS HSSC–I
(Old Curriculum 2006)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly. **(14 x 3 = 42)**

(i)	Determine the equivalent value of 1 radian in degrees.	03	OR	A bus horn produces sound of frequency 200 Hz and is receding from a stationary observer at 18 m/s. if speed of sound is 331 m/s then what frequency does the observer hear?	03
(ii)	Show that the equation for liquid pressure is $P = pgh$ dimensionally correct.	03	OR	Briefly explain the working of RADAR.	03
(iii)	The magnitude of dot and cross product of two vectors are $6\sqrt{3}$ and 6 respectively. Find the angle between the vectors.	03	OR	An oil film spreading over a wet footpath shows colors. How does it happen? Explain.	03
(iv)	Briefly explain with example, that for a body in complete equilibrium both first and second conditions of equilibrium must be satisfied.	03	OR	State the two rules of Huygens's principle.	03
(v)	Write any two applications of polarized light.	03	OR	Is it possible to convert internal energy into mechanical energy? Explain with an example.	03
(vi)	Briefly explain impulse and state how it is related to linear momentum.	03	OR	Briefly explain Isothermal process.	03
(vii)	If two balls of equal masses have elastic collision in one dimension, calculate velocities after collision.	03	OR	Briefly explain vector product of 2 vectors. Give one example.	2+1
(viii)	A ball of mass 100 g is thrown vertically upward at a speed of $25ms^{-1}$. If no energy is lost, determine the height it would reach.	03	OR	Briefly explain angular velocity and angular acceleration. Also their formulas.	1.5 + 1.5
(ix)	Write any three characteristics of waves.	03	OR	Derive an expression of velocity of a Disc when it rolls down an inclined plane.	03
(x)	A body will be weightless when the elevator falls down just like a free falling body. Explain.	03	OR	With the help of one daily life example define damped oscillations.	03
(xi)	Why the terminal speed of a parachutist after opening a parachute is less to the terminal speed before opening a parachute?	03	OR	The speed sound in air at $0°C$ is $332ms^{-1}$. What will be the speed of sound at $220°C$.	03
(xii)	Velocity of water in 6 inch diameter pipe is $5ft\,s^{-1}$. find the velocity in a 3 inch diameter pipe which connects with it.	03	OR	The length of a pendulum is $(100 \pm 0.1)cm$. If acceleration of free fall is $(9.8 \pm 0.1)ms^{-2}$. Calculate percentage in time period of the pendulum.	03
(xiii)	What is torque? Explain why it is equal to the vector product of force and moment arm.	03	OR	Does the tension in the string of a swinging pendulum do any work? Explain.	03
(xiv)	Differentiate between free and forced oscillations with example.	03	OR	Smoke rises in a chimney when a breeze is blowing. Use Bernoulli's principle to explain the formula.	03

SECTION – C (Marks 26)

Attempt the following questions.

Q.3	a. What is meant by absolute potential energy? Derive a relation for absolute potential energy of body at the surface of the earth. b. A uniform rod 1m long with weight 6 N can be supported in a horizontal position on a sharp edge with weights of 10 N and 15 N suspended from its ends. What is the position of point of balance.	4+3	OR	What is Young's double slit experiment? Discuss in detail with the help of a diagram. Also find expressions of position of fringes on screen and fringe spacing.	07
Q.4	Define centripetal force and derive its relation.	06	OR	Keeping in mind the relative velocity of source and observer discuss Doppler effect in detail also compare real and apparent frequencies.	06
Q.5	a. Explain Newton's formula for the speed of sound. Show that how it was corrected by a French scientist Laplace? b. A Carnot heat engine absorbs 2000 J of heat from the source of heat engine at 227°C and rejects 1200J of heat during each cycle to sink. Calculate efficiency of engine, temperature of sink and amount of work done during each cycle.	4+3	OR	a. What is projectile motion? Give examples. Derive mathematical equations for range of a projectile. b. Calculate the angle of projection for which K.E. at the highest point of its trajectory is equal to one-fourth of its K.E. at point of projection.	4+3
Q.6	Describe viscous drag. Also derive an expression for terminal velocity.	06	OR	Discuss the effect of temperature and density on speed of sound in air.	3+3

- $R = \frac{v_i^2 \operatorname{Sin} 2\theta}{g}$
- $H = \frac{v_i^2 \operatorname{Sin}^2 \theta}{2g}$
- $T = \frac{2v_i \operatorname{Sin} \theta}{g}$
- $f' = (\frac{v}{v+a})f$
- $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$
- $P = \frac{mgh}{t}$
- $\vec{A} \times \vec{B} = AB \operatorname{Sin}\theta \hat{n}$
- $v_t = \frac{2\rho g r^2}{9\eta}$
- $a_c = \frac{v^2}{r} = \frac{v^2}{R}$
- $F_c = \frac{mv^2}{r}$
- $T = 2\pi\sqrt{\frac{l}{g}}$
- $v_e = \sqrt{\frac{GM}{R+h}}$
- $\vec{A}.\vec{B} = AB \operatorname{Cos}\theta$
- $T = 2\pi\sqrt{\frac{m}{k}}$
- $f' = \left(\frac{v-b}{v+a}\right)f$
- $d = \frac{1}{N}$
- $V_{esc} = \sqrt{2gR}$
- $P = Fv$
- $K.E = \frac{1}{2}mv^2$
- $d \operatorname{Sin}\theta = n\lambda$
- $f' = \left(\frac{v+b}{v-a}\right)f$
- $v_t = v_0 + (0.61)t$
- $2d \operatorname{Sin}\theta = m\lambda$
- $R_e = 6.4 \times 10^6 m$
- $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$
- $\operatorname{Sin}^2\theta + \operatorname{Cos}^2\theta = 1$
- $M_e = 6 \times 10^{24} kg$
- $G = 6.673 \times 10^{-11} Nm^2 kg^{-2}$